

Shoulder — IM

NOVA-FLEX (MSC) 2 cc IM · physician treatment protocol

A standardized, physician-directed framework for shoulder pain managed by the intramuscular route: a single NOVA-FLEX dose placed in the middle deltoid of the affected side. Intended for patients who decline or are not candidates for intra-articular injection, and for practices without image guidance.

Indications

- Chronic shoulder pain > 6 months
- Rotator cuff tendinopathy or degenerative cuff pathology
- Mild-to-moderate glenohumeral osteoarthritis
- Chronic inflammatory shoulder pain
- Patients who decline or are not candidates for intra-articular injection
- Failed PT, activity modification, medication, or prior injections

Patient preparation

1. Verify indication; review imaging when available.
2. Obtain informed consent.
3. Record baseline VAS pain score and shoulder function.
4. Position seated, arm relaxed at the side, deltoid fully exposed.
5. Landmark the middle deltoid — 2–3 finger-breadths below the acromion, mid-lateral muscle belly.
6. Prepare skin with alcohol or chlorhexidine.

POST-PROCEDURE CARE

- Observe 15 minutes following injection
- Mild soreness may occur for 24–72 h
- Avoid NSAIDs and alcohol for 72 h when appropriate
- No heavy lifting with the injected arm for 24–48 h; normal activity otherwise

Contraindications

- Active local or systemic infection
- Pregnancy or breastfeeding
- Insufficient deltoid muscle mass or compromised skin at the site
- Corticosteroid injection within 4 weeks
- Systemic corticosteroid use within 30 days
- Uncontrolled bleeding disorder or significant anticoagulation risk

Injection & distribution

TOTAL VOLUME

2 cc

NOVA-FLEX (MSC) 2 cc as a single intramuscular injection into the middle deltoid of the affected side. Insert perpendicular to the skin into the mid-lateral muscle belly; aspirate before injecting; inject slowly.

Equipment: 23 G, 1–1.5 in needle; alcohol or chlorhexidine prep; gloves. No image guidance required.

FOLLOW-UP & OUTCOME MEASURES

Week 2

Safety review

Week 6

Functional progress

Week 12

Primary outcome

VAS PAIN

ROM

OVERHEAD TOLERANCE

ADLs

SLEEP / NIGHT PAIN

SATISFACTION

ADVERSE EVENTS

REGULATORY

Intended for physician-directed clinical and investigational use. Providers are responsible for compliance with applicable federal, state, facility, IRB, and professional practice requirements.

Today you received an intramuscular injection of a cell product into the deltoid muscle of your upper arm. The product used is made from donated birth tissue and is not an FDA-approved treatment. This page tells you what is normal in the days ahead, how to care for yourself, and when to call us.

DATE OF TREATMENT

SITE(S) TREATED

NEXT VISIT

WHAT TO EXPECT

Changes are gradual — do not judge the result in the first days

FIRST 72 HOURS

Soreness at the injection site — much like after a vaccine — is common and usually settles in one to three days. Sleeping on that side may be uncomfortable the first night or two.

WEEKS 1–4

Normal activity is fine as comfort allows. Ease back into heavier use of the arm over the first days.

WEEKS 4–12

Progress is assessed at your follow-up visit against your baseline pain and function scores.

DO

- Keep the arm gently moving; a sling is not needed
- Sleep on the other side for the first few nights
- Use cold packs 10–15 minutes at a time for comfort
- Take acetaminophen (Tylenol) for soreness if needed

AVOID

- Anti-inflammatory pain relievers for 72 hours
- Alcohol for 72 hours
- Heavy lifting with the injected arm for 24–48 hours
- Massage of the injection site for 48 hours

CALL US RIGHT AWAY IF

For an emergency, call 911 first

- The injection site becomes increasingly hot, red, or swollen
- New numbness, tingling, or weakness in the arm or hand
- Fever above 101 °F or shaking chills
- Severe pain not controlled by acetaminophen and cold packs

CLINIC PHONE

AFTER-HOURS PHONE

NOTES FROM YOUR VISIT